SMART INDIA HACKATHON 2025

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KRISHIBODH

- **Problem Statement ID –SIH25062**
- **Problem Statement Title-**
  Implementation of Smart Agriculture for Efficient Cultivation in Hilly Regions, GOVT Of SIKKIM
- **Theme- Agriculture , FoodTech and Rural Development**
- **PS Category- Hardware**
- **Team ID- 12**
- **Team Name: ABHIMANYU**

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KRISHIBODH

Proposed Solution

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- **CNC-powered** automation system for **polyhouse farming**
- Executes **seeding, watering,** and **monitoring** with **zero guesswork**
- **Sensor-based precision** irrigation via microcontroller logic
- Built for **enclosed environments** to neutralize **climate uncertainty**

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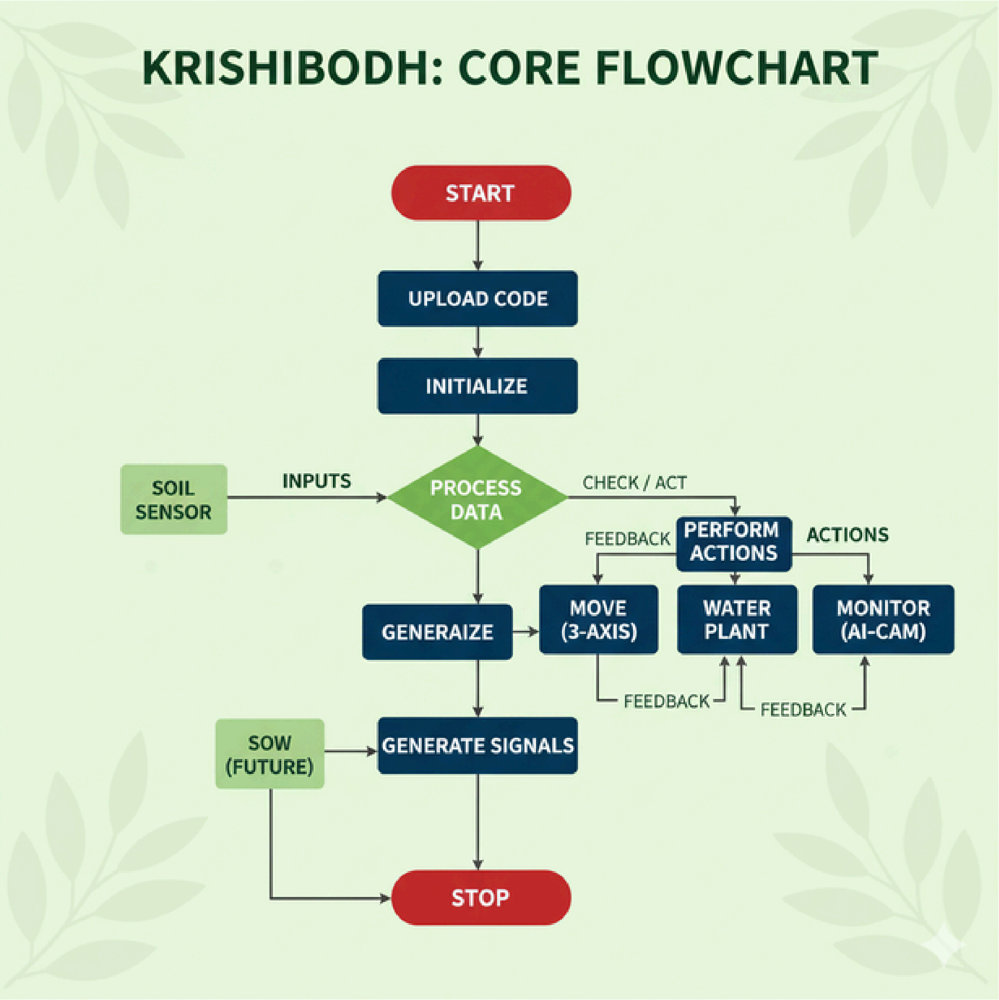
TECHNICAL APPROACH

Technologies in KRISHIBODH
Hardware components: Arduino UNO, Raspberry Pi 3, Nema 17 stepper motor, Relay module, Sensors.
Programming Languages: Python(AI/ML), C++(Microcontroller).
Frameworks & Libraries: OpenCV for image processing, Flask (Backend).
Future Enhancements: AI driven- image processing, User interface integration



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FEASIBILITY AND VIABILITY

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Aspect	Viable	Feasible
Water	Saves 40–50% water	Smart irrigation can be set up
Soil	Prevents soil erosion	Polyhouse protects soil
Technical	Works with Arduino + IoT + AI	Can be built with available tech
Economic	Increases yield & income	Low-cost, modular units

CHALLENGES	SOLUTIONS
Soil erosion during heavy monsoon rains washes away fertile topsoil	No erosion – soil is protected from rain; only controlled irrigation is used.
Water scarcity in summer terraces rely on rain-fed irrigation and stored water.	Efficient irrigation – drip + sensor-based watering saves 40–50% water.
High labour & maintenance – repairing terrace walls, bunds, and drainage systems regularly.	Low maintenance – modular automation reduces labour and upkeep.
Unpredictable rainfall dependency – crop loss if rains fail.	AI + IoT controlled environment – irrigation and climate independent of rainfall.

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IMPACT AND BENEFITS

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- **Social Impact**: Krishibodh improves quality of life by saving time and effort, especially for smallholder and women farmers in Jorethang, South Sikkim.
- **Empowering Innovation**: It promotes youth-led, tech-driven agriculture, boosting confidence and aligning with sustainable development goals.

- **Economic Impact**: Saves 40–50% water, reducing input costs and supporting sustainable farming during dry seasons.
- **Boosts Incomes**: Increases yields of cash crops like cardamom, ginger, oranges, tea, and vegetables, raising farmer incomes and reducing climate-related losses.
- **Strengthens Exports**: Enhances the "Sikkim Organic" brand, creating opportunities for premium global exports and more reliable farming systems.

- While KRISHIBODH focuses on **precision irrigation and crop monitoring**, AIF also improves **storage, supply chains, and market access**, boosting farmer incomes.
- Farmers can combine better yields from KRISHIBODH with **reduced post harvestlosses** for economic benefit.

Potential Impact On Sikkim

- **Jorethang & Low-Altitude South Sikkim**:
- **Hilly Terrain**:
- **Rainfall Pattern**:
- **Greenhouse Suitability**:

Environmental Impact:

- Promotes **Water Conservation** through rainwater harvesting and precision irrigation.
- Reduces **soil erosion and runoff** in terraced fields, protecting fragile Himalayan ecology.
- Encourages **climate-smart agriculture**, helping Sikkim adapt to rainfall variability and rising temperatures.

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RESEARCH AND REFERENCES

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- Krishibodh uses **CNC-like precision** with a **3-axis gantry system** to move tools (like water nozzles or sensors) accurately across the growing area.

1. High Investment Challenge

Building a polyhouse is expensive, costing around ₹38.80 lakh per acre, which is a significant barrier for most farmers.

2. Government & Bank Support

The NHB provides a 50% subsidy that, when combined with a bank loan, drastically reduces the farmer's debt burden.

3. Technology Drives Profit

Automated farming systems, like "Krishibodh," boost yields by 3-10 times and ensure superior crop quality, making the investment worthwhile.

4. Fast Loan Repayment

High profits from high-value crops allow the farmer to repay the remaining loan in less than one year, achieving financial sustainability quickly.

5. **National Cultivation Scale**

Across India, approximately 2 lakh hectares (about 494,210 acres) of land are dedicated to protected cultivation, including polyhouses.

6. **Karnataka's Leadership in Production**

Karnataka is a leader in the polyhouse sector, producing 1.25 million tonnes of fruits and vegetables annually from this farming method.

7. **Bengaluru's Subsidy Success**

Bengaluru Urban is the biggest beneficiary of the NHB subsidy scheme, successfully utilizing the ~50% subsidy for polyhouse installations.